

Building Software Models of Physical Objects

To describe a physical object in our everyday world, we often reference its attributes. When talking about a desk, you might describe its color, dimensions, weight, material, and so on. Some objects have attributes that apply only to them and not others. A car could be described by its number of doors, but a shirt could not. A box could be sealed or open, empty or full, but those characteristics would not apply to a block of wood. Additionally, some objects are capable of performing actions. A car can go forward, back up, and turn left or right.

To model a real-world object in code, we need to decide what data will represent that object's attributes and what operations it can perform. These two concepts are often referred to as an object's *state* and *behavior*, respectively: the state is the data that the object remembers, and the behaviors are the actions that the object can do.

State and Behavior: Light Switch Example

[Listing 2-1](#) is a software model of a standard two-position light switch written in procedural Python. This is a trivial example, but it will demonstrate state and behavior.

File: LightSwitch_Procedural.py

```
# Procedural light switch

❶ def turnOn():
    global switchIsOn
    # turn the light on
    switchIsOn = True

❷ def turnOff():
    global switchIsOn
    # turn the light off
    switchIsOn = False

# Main code
❸ switchIsOn = False # a global Boolean variable
```